What Makes Medicinal Leaf Classifiers Transfer from Studio to Field?
A Controlled Study of Leaf Location, Localised Pooling and Measured Geometry

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Abstract

Medicinal-leaf classifiers report near-perfect accuracy on studio photographs of single leaves, but how they transfer to field photographs, and why, is largely unknown. We study this question on the 30-species Medicinal Leaf Dataset with a de-duplicated, leaf-grouped protocol and test on 20 species of the DIMPSAR field photographs. Holding backbone, data and training fixed, we compare a plain classifier with a ladder of additions: pooling over the leaf mask, localised pooling (image-axis strips or object-centred sweep slabs), and *sweep-valuation attention* (SVA), a module we introduce whose tokens additionally carry the area and boundary length that a learned, appearance-weighted cubical complex of the blade and venation adds in each slab. Clean accuracy is saturated for every model. On field photographs with ConvNeXt-T (5 folds $\times$ 5 seeds, corrected resampled tests with Holm correction, and a bootstrap over runs and field species), the transfer gain decomposes into two parts of similar size: knowing where the leaf is (+7.5 points over 33.6%) and pooling locally within it (+4.6 to +6.2 more). Both parts persist without rotation augmentation and add to a strong background-compositing augmentation (best model 55%). Measured geometry adds nothing detectable on top of localised pooling: SVA (47.4%) is statistically tied with strip pooling (46.8%) and with its own slab-only variant, and fusing the same measurements globally gives no gain at all. The localisation gain is small on EfficientNet-B0 and absent on ViT-S and on the BioCLIP 2 foundation model, which reaches 81% without any module. We release the protocol, a corruption suite, cue-conflict and topology probes, all code and every run output at `https://github.com/Devguru-codes/research_paper_leaf`.

1 Introduction

Medicinal plants are identified in the field mostly from their leaves, and errors are not equally costly: confusing two culinary herbs is harmless, but confusing a remedy with a toxic look-alike, such as *Nerium oleander* with other Apocynaceae, can be dangerous [1]. Deep networks report 97.5–99.7% accuracy on Indian medicinal leaf benchmarks [2, 3]. Those numbers, however, are obtained on single leaves photographed on white backgrounds, typically with random image-level splits. They say little about how the models behave when the same species is photographed in the field, against soil, tiles or cloth, with a phone camera, which is where such a classifier would be used.

Botanists describe a leaf by its geometry: how wide the blade is along its length, where it tapers, how ragged its margin is, how its veins branch. These characters are measured *at particular places* on the leaf. Standard CNN and ViT classifiers average their features over the whole image and have no built-in notion of where the leaf is, let alone of localised geometry. A natural hypothesis is therefore that architectures which measure leaf geometry locally, and look at the image where they

measure, should transfer better from studio to field. Recent work fuses topological and morphological summaries with deep features [4–7], but usually as global descriptors, and without controls that would tell whether the geometry itself is what helps.

We test this hypothesis with a ladder of matched models that share backbone, data, vein head and training, and differ in one ingredient at a time: (1) a plain classifier; (2) the same features pooled over the leaf mask, i.e. knowing *where* the leaf is; (3) localised pooling over strips of the image or over slabs of an object-centred directional sweep; and (4) *sweep-valuation attention* (SVA), which we introduce as the most structured member of the family: for each slab of the sweep it measures how much area and boundary length a learned, appearance-weighted cubical complex of the blade and venation adds there, and lets that measurement attend to the image patches of the same slab. A global fusion of the same measurements tests whether geometry helps without localisation.

The answer is clear and partly negative. On ConvNeXt-T the studio-to-field gain over a plain model is large (14 points) but splits into two parts of similar size: knowing where the leaf is, and pooling locally within it. Measured geometry, whether fused globally or attached to localised tokens, adds nothing detectable. The gain is small on EfficientNet-B0, absent on a ViT, and irrelevant for a biology foundation model, which transfers far better on its own.

Contributions.

1. **A controlled decomposition of studio-to-field transfer** for medicinal leaves on four backbones, with 5 folds $\times$ 5 seeds for the main backbone, pre-specified contrasts, corrected resampled tests with Holm correction and a bootstrap over runs and field species. Leaf location and localised pooling each add 5–8 points on raw field photographs of ImageNet ConvNeXt-T; the gains hold without rotation augmentation and add to strong background-compositing augmentation.

2. **Sweep-valuation attention**, a differentiable, backbone-agnostic module that turns a directional filtration of appearance-weighted cubical complexes into slab tokens, and a careful negative result: its measured geometry does not improve transfer beyond localised pooling, its slab-restricted attention can be replaced by global attention, and Euler-characteristic tokens make models sensitive to topology but transfer worse.

3. **A leakage-audited protocol and probes** for the Medicinal Leaf Dataset: duplicate and reshot grouping, a corruption suite (MedLeaf-C), a leaf cue-conflict set, a perceptually matched topology-destruction test and a verified mapping of 20 DIMPSAR field species.

Code, splits, probes, statistics and the outputs of all training runs are released.

2 Related work

Medicinal leaf identification. Work on Indian medicinal leaves has moved from hand-crafted shape, colour and texture descriptors to transfer learning. DeepHerb combined Xception features with SVMs [2]; later CNN, hybrid CNN–transformer and feature-fusion models report 98.9–99.7% on this and related datasets [3], typically with random image-level splits. Closest to our biological motivation, a dual-branch network fuses an RGB branch with a branch on skeletonised veins by late concatenation [8]. A cascaded taxonomic classifier targets open-world medicinal plants [9]. We reproduce DeepHerb-style and dual-branch pipelines under our leakage-audited protocol.

Studio-to-field shift in plant recognition. The gap between curated and field images is well documented for plant disease recognition: a model trained on the laboratory images of PlantVillage reached 99.4% on held-out laboratory images but about 31% on images taken under other conditions [10], and the background alone of those images carries enough signal to classify them [11]. For medicinal leaves, field evaluation is rare. We measure this gap for a medicinal leaf benchmark under a controlled protocol and ask which architectural ingredients close it.

Euler characteristic transforms. The ECT and the persistent homology transform encode a shape through sublevel-set topology along all directions [12]. DECT makes the ECT differentiable by sigmoid relaxation and learns directions end-to-end [13]. WECT weights cells by image intensity [14]. Blaser et al. encode the ECT as per-direction token sequences, including convolutional and complex-valued variants that are equivariant to planar rotations [7]. For plants, Ayub et al. show that ECT images of *blade contours* classify complex leaf shapes and even predict venation from blade geometry [15]. Their ECT is computed on the outline only; venation is a prediction target, and appearance is not used. We build on DECT's relaxation and on these encodings. We do not claim weighted, tokenised or rotation-equivariant ECTs, or ECTs of leaves, as contributions.

Topology meets attention. Topological summaries are increasingly fused with deep features. PHG-Net injects persistence-diagram features into CNN/transformer classifiers [4]; TDA-ViT fuses topological and ViT features [5]; GraPHFormer aligns graph topology and vision representations [16]. Gu et al. process multi-filtration cubical persistence diagrams with cross-attention and fuse the embedding with CNN/transformer features [6]. TopoFormer slices graphs along filtrations into attention-friendly topological token sequences [17]. For time series, Faghihi and Saki add persistent-homology and directional Euler statistics of each input window as additive biases on attention logits [18]; the topology summarises the whole window rather than defining where each query may attend. In these methods the topological signal enters as a *global* descriptor or as tokens among themselves, so where the visual model looks is not tied to where the topology was measured. SVA differs in exactly this respect: each filtration slab yields both a valuation increment and the spatial support of that token's attention. Our controls (global fusion of the same valuations; slab attention without valuations; mask and strip pooling) separate these ingredients, and we find that the localisation, not the measured topology or geometry, carries the effect.

Intrinsic volumes and directional pooling. Area, boundary length and Euler characteristic (the planar intrinsic volumes, or Minkowski functionals) are classical morphological descriptors with efficient digital estimators [19, 20], and the perimeter of a planar shape can be recovered from its ECT [21]; by Hadwiger's theorem they span all continuous motion-invariant valuations [22, 23]. We use them as filtration increments along a sweep, not as global descriptors, and do not claim them as new features. Strip pooling [24] and axial attention aggregate features along image rows and columns. Our strip control instantiates this family; unlike SVA, its strips are fixed to the image axes, carry no measured geometry and are not defined by a filtration of the object.

Taxonomy and uncertainty. Hyperbolic and hierarchy-aware embeddings have been applied to plant and biological taxonomies [25, 26], and BioCLIP 2 learns hierarchical contrastive representations at scale [27]. Conformal prediction for long-tailed plant identification [28] and for hierarchical labels [29, 30] is established; we do not address uncertainty quantification here, because the calibration data for toxic species in this dataset are too few (about ten images per fold) for meaningful hazard-specific guarantees.

Shape versus texture. ImageNet CNNs are biased towards texture [31]. We adapt the cue-conflict paradigm to leaves (silhouette of one species filled with the lamina of another) and add a topology-destruction probe, in order to test what SV-Former actually relies on.

3 Data and leakage-audited protocol

Dataset. We use the segmented subset of the Medicinal Leaf Dataset of Roopashree and Anitha [32] (CC BY 4.0): 1,835 RGB photographs (1600×1200) of single leaves from 30 Indian medicinal species on a white background, with 34–122 images per species ($3.6\times$ imbalance). The leaf covers 4.4–68.5% of the frame (median 34.8%). The species span 28 genera, 19 families and 12 orders of the APG IV system, with genuine sibling groups (e.g. Apocynaceae: *Nerium*, *Tabernaemontana*, *Carissa*; Myrtaceae: *Psidium* and two *Syzygium*; Moraceae: two *Ficus* and *Artocarpus*; Lamiaceae: *Mentha*, *Ocimum*, *Plectranthus*); see table 6.

Audit. Reported accuracies on this dataset are near saturation (97.5–99.7% [2, 3]) and use random image-level splits. Our audit found three issues. (i) *Exact duplicates*: 10 pairs of identical photographs saved under different names, all in *Carissa carandas* (1,825 images remain after removal). (ii) *Re-shots of the same physical leaf*: some leaves were photographed more than once, at times under different illumination or orientation. Hashing alone cannot separate these from merely similar leaves (e.g. lanceolate *Alpinia* silhouettes), so we link two images when their raw-frame 1024-bit dHash distance is ≤ 20, or when SIFT matching with RANSAC yields ≥ 40 geometrically consistent inliers whose similarity transform has scale in $[0.8, 1.25]$ and whose convex hull covers $\geq 25\%$ of the leaf. These criteria were set by visual inspection and reject hub matches caused by repetitive texture (e.g. *Mentha*). The resulting 22 multi-image groups (45 images, at most 3 per group) are kept intact across splits. Over-grouping is conservative: it constrains splits but cannot leak. (iii) *Filename collisions*: *Psidium guajava* and *Punica granatum* share the `PG-S-xxx` prefix (65 colliding names), so files must be indexed by class and name.

Protocol. All results use stratified, group-aware 5-fold cross-validation on the 1,825 de-duplicated images. Inside each training part, one group-aware stratified holdout ($\approx 1/6$) is used for monitoring only. Epoch count and hyper-parameters are fixed in advance, with no model selection on test or calibration data. With random image-level splits, 1.4–3.0% of test images per fold have a duplicate or re-shot in the training part (section 6.8).

Preprocessing. The leaf mask is obtained by thresholding the white background (all channels < 235) followed by morphological cleaning. Each image is cropped to a square centred on the leaf whose inscribed circle contains the whole leaf, so that any rotation about the centre is lossless, and resized to 448^2 (training input 384^2). Vein pseudo-labels are multiscale Sato ridge responses on CLAHE-normalised luminance inside the eroded mask.

Cross-domain test set. For transfer to field photographs we use DIMPSAR [33] (CC BY 4.0), whose single-leaf subset was captured with smartphones against varied backgrounds (tiles, soil, cloth, paper). Vernacular folder names were mapped to species and checked visually. Twenty species overlap ours; note that Kannada *Arali* denotes *Ficus religiosa* (peepal), not *Nerium oleander*. Leaves are localised with our own masker, a ConvNeXt-T FPN trained only on our leaves composited onto DTD textures [34] and procedural backgrounds; no third-party segmentation model is used.

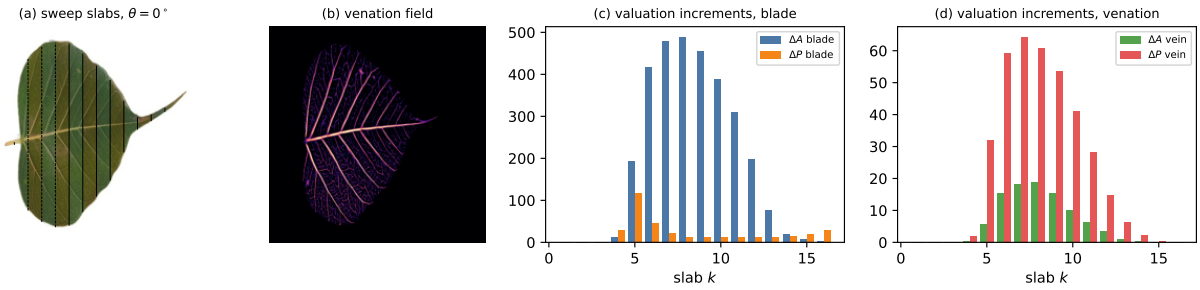


Figure 1: The sweep-valuation signal for one leaf (*Ficus religiosa*) and one of the K=32 directions. (a) The leaf is partitioned into T=16 slabs along θ in its centroid/RMS frame. (b) Ridge-filter vein pseudo-label that supervises the learned venation field. (c,d) Area and perimeter increments per slab for the blade and the venation field, shown here with unit weights; in SVA the cell weights are appearance-conditioned. Each slab's increments form the query of one token, which attends to the image patches of the same slab. The long perimeter in the last blade slabs comes from the drip tip.

4 Models: sweep-valuation attention and matched controls

Figure 1 summarises SV-Former. An appearance backbone (CNN or ViT) produces patch tokens. Two structural fields are derived from the same image: the *blade*, a deterministic leaf mask, and a *learned venation field*, a vein-probability map predicted by a light head on early backbone features. A directional sweep partitions the leaf into ordered slabs. For every direction–slab pair, SVA measures how much of each field's *intrinsic volumes* (area and boundary length) enters the filtration inside that slab, and uses *the same slab* to restrict where the resulting token attends among the patch tokens.

4.1 Weighted cubical complexes

The leaf crop is resampled to a $G \times G$ grid (G=96). Pixel centres x_p are expressed in a frame centred at the blade centroid and scaled by its root-mean-square radius, which makes all quantities below invariant to translation and isotropic scale. We use the cubical complex whose 0-cells are pixels, whose 1-cells are pairs of 4-adjacent pixels and whose 2-cells are 2×2 pixel squares. Given per-pixel weights $a(p) \in [0, 1]$, cells inherit the minimum weight of their vertices,

$$w_0(p) = a(p), \qquad w_1(e) = \min_{v \in e} a(v), \qquad w_2(q) = \min_{v \in q} a(v). \tag{1}$$

This keeps face monotonicity $w(\text{coface}) \le w(\text{face})$, so the weighted complex is a fuzzy subcomplex. For binary a it reduces exactly to the standard cubical complex of the set.

Appearance-conditioned weights. Two weight fields are built from mid-level backbone features f (bilinearly sampled at the grid), a small per-pixel MLP g and a sigmoid σ:

$$a_{\mathrm{b}}(p) = m(p)\,\sigma(g_{\mathrm{b}}(f(p))), \qquad a_{\mathrm{v}}(p) = m(p)\,v(p)\,\sigma(g_{\mathrm{v}}(f(p))), \tag{2}$$

where m is the blade mask and v is the predicted vein probability. The last layer of g is initialised so that $a \approx 0.88\,m$ at the start of training. The vein head is pseudo-supervised by a multiscale Sato ridge response [35] inside the eroded mask and also receives task gradients through SVA. In contrast to fixed intensity weights (WECT [14]), the weights are learned and conditioned on appearance. We stress that a_{v} is a *learned* venation field, not an observed vascular structure.

4.2 Valuations of a weighted cubical complex

Let $H_d = \sum_{c \in C_d} w_d(c)$ be the weighted number of d-cells. For a binary set the three quantities

$$A = H_2, \qquad P = 2H_1 - 4H_2, \qquad \chi = H_0 - H_1 + H_2 \tag{3}$$

are the area (number of unit squares), the boundary length in pixel units (every square contributes four edges, and an edge shared by two squares is interior) and the Euler characteristic. In the continuum, area, perimeter and Euler characteristic are the three planar intrinsic volumes, and Hadwiger's theorem [22, 23] states that every continuous, rigid-motion-invariant valuation on planar *convex* bodies is a linear combination of them. We use the theorem only as a motivation for this choice of basis and make no claim that it carries over to our setting, for three reasons. Leaves are not convex. The digital perimeter $2H_1 - 4H_2$ is the ℓ_1 (city-block) length of the boundary, which is not rotation invariant: it overestimates the Euclidean length of a straight boundary at angle α to the grid by the factor $|\cos\alpha| + |\sin\alpha|$ (up to $\sqrt{2}$), and of a disk by $4/\pi \approx 1.27$ on average [20]. Finally, (3) is applied to the fuzzy complex (1) by linearity in the cell weights; the result is a weighted count, not a geometric valuation of any set. Our proposed token uses the area and perimeter counts. The Euler characteristic is retained as an ablation (section 6.6).

4.3 Filtration increments along a sweep

For a direction $\theta \in S^1$ we use the lower-star filtration, $h_\theta(v) = \langle x_v, \theta \rangle$ and $h_\theta(c) = \max_{v \in c} h_\theta(v)$. Following the sigmoid relaxation of DECT [13], a cell enters the filtration at threshold t with soft indicator $\sigma(\lambda(t - h_\theta(c)))$. Fix thresholds $t_0 < t_1 < \cdots < t_T$, where $t_1, \ldots, t_{T-1}$ are uniform on $(-R, R)$, $t_0 = -\infty$ and $t_T = +\infty$. The soft membership of height h in slab k is

$$s_k(h) = \sigma\big(\lambda(t_k - h)\big) - \sigma\big(\lambda(t_{k-1} - h)\big), \qquad k = 1, \ldots, T, \tag{4}$$

and because the outer thresholds are infinite, $\sum_k s_k(h) = 1$ for every h. The *filtration increment* of the weighted d-cell count in slab k is $\Delta H_d(\theta, k) = \sum_{c \in C_d} w_d(c)\, s_k(h_\theta(c))$, and the valuation increments follow by linearity:

$$\Delta A(\theta, k) = \Delta H_2, \qquad \Delta P(\theta, k) = 2\Delta H_1 - 4\Delta H_2, \qquad \Delta\chi(\theta, k) = \Delta H_0 - \Delta H_1 + \Delta H_2. \tag{5}$$

Each is the change of the corresponding relaxed count between t_{k-1} and t_k, and $\sum_k$ telescopes to the total valuation of the leaf, independently of θ. We compute (5) for the blade (a_b) and the venation field (a_v) with K uniform directions and T slabs (K=32, T=16, R=2, $\lambda = \kappa T/2R$ with κ=4).

Sweep-valuation token. The structural feature of token (θ, k) is the 8-vector of area and perimeter increments of both fields together with their cumulative values up to slab k,

$$\phi(\theta, k) = \mathrm{slog}\Big[\Delta A_\mathrm{b}, \Delta P_\mathrm{b}, \Delta A_\mathrm{v}, \Delta P_\mathrm{v},\ \textstyle\sum_{j \le k}(\cdot)\Big], \qquad \mathrm{slog}(x) = \mathrm{sign}(x)\log(1 + |x|). \tag{6}$$

The increments describe the local width profile and boundary complexity of the slab; the cumulative values locate it within the leaf. The Euler variant replaces ϕ by $[\Delta\chi_\mathrm{b}, \Delta\chi_\mathrm{v}, \chi_\mathrm{b}(t_k), \chi_\mathrm{v}(t_k)]$, and the Hadwiger variant uses all three valuations (12 features).

Efficient computation. Directions are fixed and coordinates come from the mask, so the heights carry no gradient; only the weights do. We hard-bin heights into n_b=256 fine bins, scatter-add cell weights per dimension into a per-direction histogram, and apply the slab kernel (4) evaluated at bin centres with one matrix product (the kernel is built in double precision). This costs $O(BKN)$ memory for N cells, needs no sorting, and is exact in the telescoping and sharp limits (verified by unit tests, section A).

4.4 Slab-restricted attention

Each direction–slab pair (θ, k) defines one sweep token with query $q_{\theta k} = \mathrm{MLP}(\phi(\theta, k)) + e_k$, where e_k is a learned slab-index embedding. The token cross-attends to the backbone patch tokens $\{z_p\}$ with an additive bias derived from the *same* soft slab membership (4), evaluated at patch centres:

$$\mathrm{attn}_{(\theta,k)}(p) \propto \exp\Big(\tfrac{\langle W_Q q_{\theta k}, W_K z_p\rangle}{\sqrt{d}} + \log\big(s_k(\langle x_p, \theta\rangle)\,\bar{m}(p) + \epsilon\big)\Big), \tag{7}$$

where $\bar{m}(p)$ is the fraction of leaf pixels in patch p and $\epsilon = 10^{-4}$. The structural measurement and the visual evidence attached to it are thus taken over the same filtration-defined region. To the best of our knowledge, we did not find prior work that explicitly co-registers filtration-derived valuation increments with attention restricted to the corresponding filtration slab (section 2). We treat this restriction as a design choice and test it directly: in our experiments it can be replaced by global attention without loss (section 6.6). The attention output is added to the query with a mass gate $M/(M + 0.5)$, $M = \sum_p s_k \bar{m}$, so that slabs outside the leaf contribute little. The KT=512 tokens are mixed by a two-layer *sweep transformer* whose learned relative position bias depends only on $(\Delta\theta \bmod K, \Delta k)$. Tokens are mean-pooled and concatenated with the backbone's global token for classification. Training minimises cross-entropy (label smoothing 0.1) plus 0.2× the masked BCE of the vein head.

Rotation. For unit weights, rotating the input by a multiple of 90° cyclically shifts all increments by $K/4$ directions (section A); the appearance-weighted signal and the full model inherit the rotation behaviour of the backbone and are evaluated empirically. Rotation-equivariant ECT tokenisation as such is not our contribution [7].

4.5 A ladder of matched models

To isolate what matters, we compare six models that share the backbone, the vein head and its auxiliary loss, the training recipe and, where applicable, the token width (192) and transformer depth (2). Each step adds one ingredient; all but the first concatenate their output with the backbone's global token.

1. **plain**: backbone global token only; features are averaged over the whole image, background included;

2. **mask pooling**: adds the mask-weighted mean of the patch tokens over the leaf (an MLP per token, then masked averaging); the model knows *where* the leaf is but pools it as a whole;

3. **strip pooling** [24]: mask-weighted means of patch tokens over every row and column of the patch grid, with learned positions and the transformer; localised pooling along the image axes;

4. **slab-only**: identical to SVA except that $\phi \equiv 0$, so each query is only the slab embedding e_k; localised pooling over the object-centred sweep slabs, without measurements;

5. **SVA**: slab tokens that also carry the measured valuations;

6. **global fusion**: the full $K \times T \times 8$ valuation grid of SVA flattened through an MLP and fused globally (in the spirit of [4, 6]); the same measurements without localised pooling.

The pre-specified contrasts are: 2 vs. 1 (*leaf location*); 3, 4 and 5 vs. 2 (*localisation*); 5 vs. 4 and 5 vs. 3 (*measured geometry*, and object-centred slabs vs. image-axis strips); and 6 vs. 1 (*geometry without localisation*). On field photographs the leaf mask of models 2–6 comes from our masker (section 3); the plain model receives no mask, which is exactly the difference contrast 2 vs. 1 measures.

5 Experimental setup

Backbones and training. We use two ImageNet-1k CNNs (ConvNeXt-T and EfficientNet-B0), ViT-S/16 (AugReg, ImageNet-21k→1k) and the BioCLIP 2 ViT-L/14 visual tower (all but the last two blocks frozen). All models of a backbone share the recipe: AdamW (new modules lr 10^{-3}, weight decay 0.05), 2 warm-up epochs then cosine decay, batch 32, mixed precision, label smoothing 0.1. The backbone lr is 10^{-4} for ConvNeXt-T, 2×10^{-4} for EfficientNet-B0 and 5×10^{-5} for ViT-S, all trained 40 epochs at 384^2; BioCLIP 2 uses lr 3×10^{-5}, 20 epochs and 224^2 input. The recipe was fixed in advance and not tuned per model. Augmentation uses uniform rotation in $[-180°, 180°]$, scale $[0.85, 1.15]$, translation $\pm 5\%$, horizontal flips and mild colour jitter, applied jointly to image, mask and vein pseudo-label. No early stopping or test-time augmentation is used. Runs use Kaggle T4 GPUs.

Runs. All splits are grouped stratified 5-fold cross-validation over the de-duplicated, leaf-grouped data (section 3). For ConvNeXt-T, the six models of section 4.5 are trained with 5 seeds per fold (25 runs each); the field test and MedLeaf-C are run on all seeds, the other probes on seed 0. EfficientNet-B0 uses 3 seeds per fold, ViT-S and BioCLIP 2 one seed per fold (mask pooling only on ConvNeXt-T). Two further ConvNeXt-T arms test whether the effects depend on augmentation. *Without rotation augmentation*: plain, strip, slab-only and SVA, 5 folds × 5 seeds. *Strong augmentation*: plain, strip and SVA, 5 folds × 3 seeds, with background compositing (with probability 0.8 the white background is replaced by a random procedural texture, blended through the slightly eroded mask), Gaussian blur ($p = 0.3$) and Gaussian noise ($p = 0.3$) on top of the standard augmentation.

Statistics. The primary outcome is 20-way accuracy on raw DIMPSAR field photographs; the masked photographs are the secondary outcome. The contrasts of section 4.5 were fixed before the mask-pooling runs were analysed. Each contrast is the mean difference over matched fold–seed runs. We give two 95% intervals: the Nadeau–Bengio corrected resampled t-test [36] over runs (test/train ratio 0.3, accounting for overlapping training sets), which captures training variability, and a two-way bootstrap that resamples runs and field species (clusters of images) jointly, which also captures sampling of the fixed field test set. p-values come from the corrected test and are Holm-corrected [37] over all 14 contrasts (7 contrasts × 2 photo types). Secondary comparisons (other arms, backbones, probes) are reported with the corrected interval and win counts and without multiplicity claims.

Ablations of SVA. Euler-only and full-Hadwiger tokens; blade-only and vein-only fields; unit cell weights; *permuted* learned weights (the same weight values shuffled within the leaf before computing the valuations); global instead of slab-restricted attention; a coarser sweep (K=16, T=8); and a softer or sharper relaxation ($\kappa \in \{1, 16\}$). All ablations run on seed 0, folds 0–2; the four most informative (unit weights, global attention, coarse sweep, Euler token) are repeated on all five folds and seeds 0–2 and compared with SVA on the same runs.

Probes. All probes are applied to each fold's test images with fixed seeds.

- **Field photographs**: models trained on the studio images are tested on the 2,000 DIMPSAR photographs of the 20 overlapping species (section C), on raw crops with the background kept and on crops whose background was whitened by our masker. Accuracy is 20-way, i.e. restricted to the shared species; this uses the knowledge of which species can occur and is the same for all models.
- **MedLeaf-C**: ten corruptions × three severities: Gaussian noise, blur, JPEG, low resolution, darkening, contrast loss, hue shift (yellowing), desaturation (severity 3 is grayscale), occluders, and pasted procedural/soil-like backgrounds. Masks are the oracle masks transformed consistently.
- **Rotation sweep**: accuracy at $0°, 30°, \ldots, 330°$.
- **Cue conflict**: the silhouette of a test leaf A is filled with the lamina of a test leaf B of another species. B is aligned by principal axes and enlarged until it covers A, so that B's outline contributes no shape cue. Shape bias $= n_A/(n_A + n_B)$ where n_A (n_B) counts predictions of A's (B's) species.
- **Topology destruction**: three interior holes in the blade (exact $\Delta\chi = -3$, verified) versus three boundary bites of equal removed area ($\Delta\chi = 0$, verified), and inpainting on strong veins versus vein-free lamina. Every edit records LPIPS, pixel RMSE and a colour-histogram distance; the response (KL divergence between clean and edited predictions) is compared within LPIPS strata (section 6.7).

6 Results

All numbers below are generated from the run outputs by the released scripts; ± is the standard deviation over runs. Differences are given as mean over matched fold–seed runs, with the number of runs in which the first model wins and 95% intervals as described in section 5.

6.1 In-domain accuracy is saturated

On the de-duplicated, grouped protocol every model on every backbone reaches 99.9–100% clean accuracy and $\geq$ 98.5% accuracy under any rotation (table 1). The studio benchmark therefore cannot separate architectures, and none of the added modules costs clean accuracy. All further results concern robustness and, primarily, transfer to field photographs.

Table 1: Main results (%). Clean = in-domain grouped 5-fold accuracy; Rot. min = worst accuracy over 12 rotations; MedLeaf-C = mean over 10 corruptions × 3 severities; Shape bias from the cue-conflict set; Field = 20-way accuracy on DIMPSAR field photographs (auto-masked or raw). ConvNeXt-T: 5 folds × 5 seeds (MedLeaf-C: seeds 0, 3, 4, all seeds for mask pooling; Rot. min and shape bias: seed 0); EfficientNet-B0: 5 folds × 3 seeds; ViT-S and BioCLIP 2: 5 folds, one seed. Mask pooling was run on ConvNeXt-T only.

Backbone	Model	Clean	Rot. min	MedLeaf-C	Shape bias	Field (mask)	Field (raw)
ConvNeXt-T	Plain	99.9 ± 0.1	99.9 ± 0.2	82.1 ± 1.1	6.5 ± 1.5	35.2 ± 1.1	33.6 ± 1.2
	Global fusion	99.9 ± 0.1	99.7 ± 0.3	82.2 ± 1.1	7.0 ± 1.9	34.6 ± 1.5	33.5 ± 1.1
	Mask pooling	100.0 ± 0.1	–	84.8 ± 0.9	–	40.9 ± 1.3	41.1 ± 1.6
	Strip pooling	100.0 ± 0.0	99.9 ± 0.1	87.5 ± 1.0	9.1 ± 1.5	42.4 ± 1.6	46.8 ± 2.2
	Slab-only	100.0 ± 0.1	100.0 ± 0.0	88.8 ± 1.1	7.4 ± 1.9	42.1 ± 1.2	45.7 ± 1.8
	SVA (ours)	100.0 ± 0.1	99.9 ± 0.1	87.0 ± 1.4	9.0 ± 2.3	42.5 ± 2.6	47.4 ± 3.7
EffNet-B0	Plain	99.9 ± 0.1	99.9 ± 0.1	77.9 ± 1.4	–	32.2 ± 1.6	26.8 ± 3.2
	Global fusion	100.0 ± 0.1	99.7 ± 0.2	78.5 ± 0.9	–	31.7 ± 1.1	29.2 ± 2.1
	Strip pooling	99.9 ± 0.1	99.8 ± 0.1	77.2 ± 0.8	–	31.1 ± 1.1	29.1 ± 2.3
	Slab-only	99.9 ± 0.1	99.7 ± 0.2	77.8 ± 0.9	–	30.3 ± 1.4	28.3 ± 2.7
	SVA (ours)	99.9 ± 0.2	99.8 ± 0.2	78.5 ± 1.3	–	31.1 ± 1.4	29.8 ± 2.2
ViT-S/16	Plain	99.9 ± 0.1	99.9 ± 0.1	96.4 ± 0.3	9.7 ± 1.0	51.6 ± 0.4	60.4 ± 0.9
	Global fusion	100.0 ± 0.0	99.9 ± 0.1	96.2 ± 0.6	9.1 ± 1.3	50.1 ± 0.7	58.6 ± 1.8
	Strip pooling	99.9 ± 0.1	99.9 ± 0.1	96.2 ± 0.5	10.3 ± 1.3	51.2 ± 1.1	62.7 ± 1.0
	Slab-only	100.0 ± 0.0	99.8 ± 0.1	96.6 ± 0.5	11.0 ± 1.4	52.0 ± 0.8	64.8 ± 2.1
	SVA (ours)	99.9 ± 0.1	99.9 ± 0.1	96.6 ± 0.3	9.3 ± 1.8	51.9 ± 1.0	60.7 ± 1.4
BioCLIP 2	Plain	99.9 ± 0.1	99.0 ± 0.4	93.8 ± 0.5	14.0 ± 1.3	58.9 ± 0.5	81.4 ± 0.7
	Global fusion	99.9 ± 0.1	99.0 ± 0.4	93.8 ± 0.6	14.3 ± 2.3	59.7 ± 1.1	81.3 ± 0.2
	Strip pooling	99.9 ± 0.1	99.2 ± 0.5	93.8 ± 0.2	14.2 ± 1.7	58.9 ± 0.7	81.5 ± 0.5
	Slab-only	100.0 ± 0.0	98.8 ± 0.5	93.9 ± 0.6	13.9 ± 1.2	60.7 ± 0.7	81.8 ± 1.4
	SVA (ours)	100.0 ± 0.0	98.5 ± 0.5	93.5 ± 0.5	13.7 ± 2.0	59.8 ± 1.1	81.7 ± 0.8

6.2 Decomposing the studio-to-field gain

Figure 2 and table 2 show the main result. A plain ConvNeXt-T trained on studio images reaches 33.6% on raw field photographs. The localised models reach 45.7–47.4%, a gain of 12–14 points. The ladder of matched models splits this gain into two parts of similar size.

Leaf location. Pooling the same patch features over the leaf mask raises raw field accuracy to 41.1% (+7.5 points, 25/25 runs, Holm $p < 10^{-4}$), and masked-photo accuracy by +5.7. The plain model averages features over the whole photograph, and on field photographs the background (tiles, soil, cloth) is a large part of it; a model that knows where the leaf is can ignore it.

Localisation. Pooling locally within the leaf adds a further 4.6–6.2 points on raw photographs: strip pooling +5.7 (25/25, Holm $p = 0.002$), slab-only attention +4.6 (24/25, Holm $p = 0.008$) and SVA +6.2 (23/25; Holm $p = 0.18$ because its run-to-run variance is the largest). The two-way bootstrap over runs and field species excludes zero for all three. On masked photographs, where the background is already removed, localisation adds only 1.2–1.6 points (not significant).

Measured geometry. Adding the measured valuations to localised tokens does not help detectably. SVA is tied with strip pooling (+0.5, 12/25) and with its own slab-only variant (+1.7, 16/25), with intervals of roughly ±4 points around these values. Fusing the same valuations globally, without localisation, gives no gain at all over plain (−0.1, 13/25). Object-centred sweep slabs and image-axis strips perform the same.

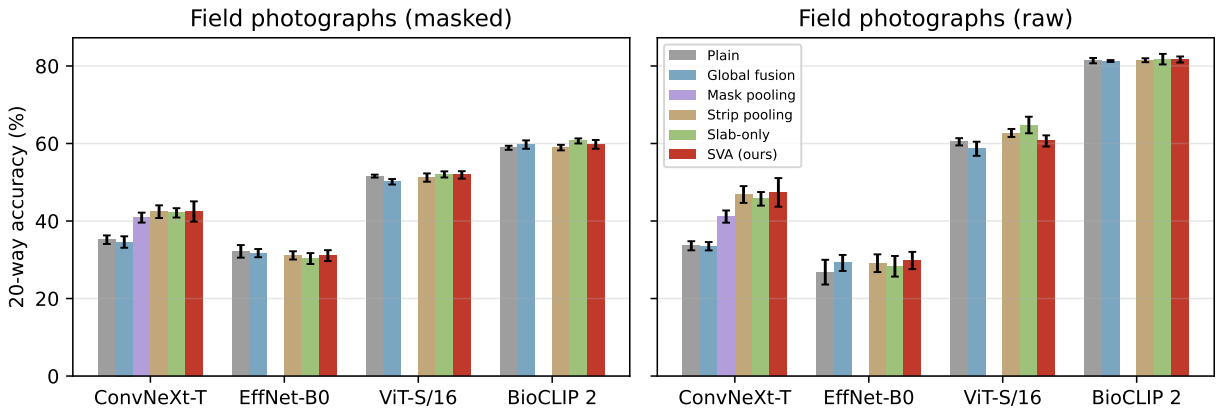


Figure 2: Studio-to-field transfer: 20-way accuracy on DIMPSAR photographs with the background whitened by our masker (left) or kept (right). Error bars: std over runs. Mask pooling was run on ConvNeXt-T only.

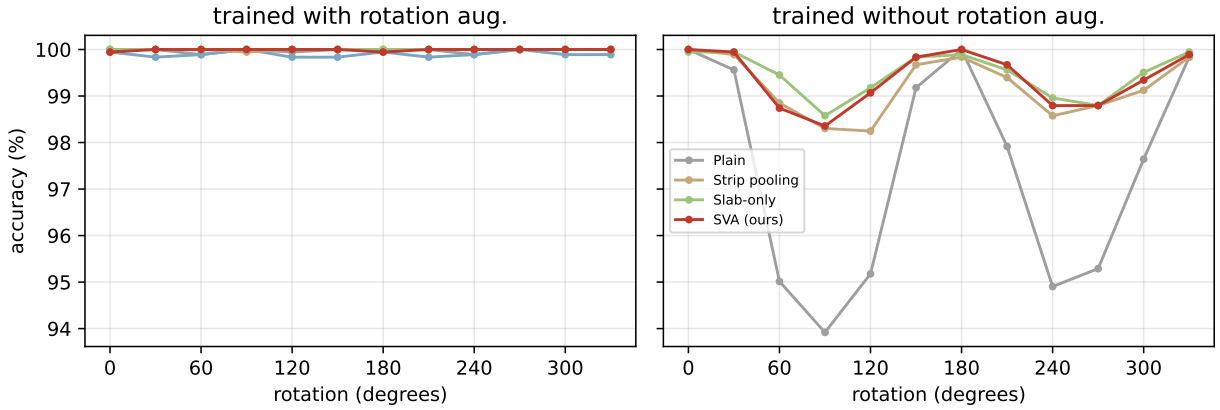


Figure 3: Accuracy versus test-time rotation (ConvNeXt-T, seed 0, mean over 5 folds), trained with (left) and without (right) rotation augmentation.

6.3 The gains are not an artefact of augmentation

Without rotation augmentation (5 folds × 5 seeds), the localised models remain far more rotation-robust than plain (worst-case accuracy 97.5–98.4% vs. 93.2%; fig. 3) and keep their field gain: plain 32.1%, slab-only 43.2%, strip 43.9%, SVA 45.5% on raw photographs (SVA vs. plain +13.5 [+9.7, +17.3], 25/25). SVA is again not distinguishable from strip pooling (+1.6, 16/25) or slab-only attention (+2.4, 20/25). With a single seed these leads had looked much larger (+5.8 and +5.2), which illustrates why we rely on many seeds.

With strong augmentation (background compositing, blur, noise; table 3), every model improves: plain rises from 33.6% to 42.0% on raw photographs, about as much as mask pooling gives without it, which is consistent with both teaching the model to discount the background. Localisation still adds on top: strip pooling reaches 53.4% and SVA 55.2%, the best field accuracy of any ImageNet model here (SVA vs. plain +13.2 [+9.9, +16.5], 15/15; vs. strip +1.9, 13/15, n.s.). The architectural and data-side remedies are therefore complementary.

Table 2: Pre-specified contrasts on field photographs (ConvNeXt-T, 25 matched runs). Δ in accuracy points with runs won; 95% intervals from the corrected resampled t-test over runs and from a bootstrap over runs and field species; p Holm-corrected over all 14 contrasts.

Photos	Contrast	Δ (points)	95% CI, runs	95% CI, runs+species	Holm p
raw	mask pooling − plain	+7.5 (25/25)	[+5.3, +9.8]	[+3.0, +12.2]	$< 10^{-4}$
	strip − mask pooling	+5.7 (25/25)	[+3.0, +8.4]	[+2.3, +9.2]	0.002
	slab-only − mask pooling	+4.6 (24/25)	[+2.1, +7.0]	[+0.7, +8.9]	0.008
	SVA − mask pooling	+6.2 (23/25)	[+1.2, +11.3]	[+2.4, +10.2]	0.18
	SVA − slab-only	+1.7 (16/25)	[-3.5, +6.8]	[-2.8, +6.5]	1.00
	SVA − strip	+0.5 (12/25)	[-3.6, +4.7]	[-3.0, +4.5]	1.00
	global fusion − plain	-0.1 (13/25)	[-1.5, +1.3]	[-2.3, +2.4]	1.00
masked	mask pooling − plain	+5.7 (25/25)	[+3.6, +7.8]	[+2.3, +9.3]	0.000
	strip − mask pooling	+1.5 (21/25)	[-0.7, +3.7]	[-1.4, +4.3]	1.00
	slab-only − mask pooling	+1.2 (20/25)	[-0.7, +3.2]	[-2.4, +5.0]	1.00
	SVA − mask pooling	+1.6 (17/25)	[-1.7, +4.9]	[-1.7, +4.9]	1.00
	SVA − slab-only	+0.4 (15/25)	[-3.2, +3.9]	[-3.4, +4.4]	1.00
	SVA − strip	+0.1 (15/25)	[-3.4, +3.5]	[-2.9, +3.4]	1.00
	global fusion − plain	-0.6 (8/25)	[-2.4, +1.2]	[-3.4, +2.3]	1.00

Table 3: Standard vs. strong (background compositing + blur + noise) augmentation, ConvNeXt-T, 5 folds × 3 seeds, %.

Augmentation	Model	MedLeaf-C	Field (mask)	Field (raw)
standard	Plain	81.1 ± 1.1	35.1 ± 1.3	33.5 ± 1.2
	Strip pooling	87.6 ± 0.8	42.9 ± 1.6	47.8 ± 2.0
	SVA (ours)	88.6 ± 0.5	43.1 ± 2.8	49.3 ± 2.8
strong	Plain	96.5 ± 0.3	43.1 ± 1.3	42.0 ± 1.7
	Strip pooling	97.9 ± 0.1	49.3 ± 1.3	53.4 ± 1.7
	SVA (ours)	97.8 ± 0.3	49.4 ± 1.7	55.2 ± 2.0

6.4 The gains depend on the backbone

On EfficientNet-B0 (3 seeds) all four structural models gain only 1.5–3 points on raw photographs (plain 26.8%, SVA 29.8%, +3.0 [−2.6, +8.6], 11/15) and nothing on masked ones. On ViT-S, localised pooling may still help (strip 62.7%, slab-only 64.8% vs. plain 60.4%, one seed), but SVA does not (60.7%). BioCLIP 2 reaches 81.4% on raw photographs without any module, about 30 points above the best ImageNet ConvNeXt-T model, and all variants lie within 0.5 points of each other. A biology foundation model thus closes most of the studio-to-field gap on its own. For both ViTs, masked photographs are 8–22 points *harder* than raw ones, whereas for the CNNs the two differ by at most about five points; the whitened background and the masker's errors apparently remove context these models use, so we treat raw photographs as the primary field test.

6.5 Corruptions and cue conflict

On MedLeaf-C (fig. 4), the ladder repeats on ConvNeXt-T: mask pooling improves on plain by +2.7 points (84.8% vs. 82.1%, 15/15), and localised pooling adds a further 2.2–4.0 (strip 87.5%, SVA 87.0%, slab-only 88.8%). The gains come from noise, blur, JPEG and low-resolution corruptions,

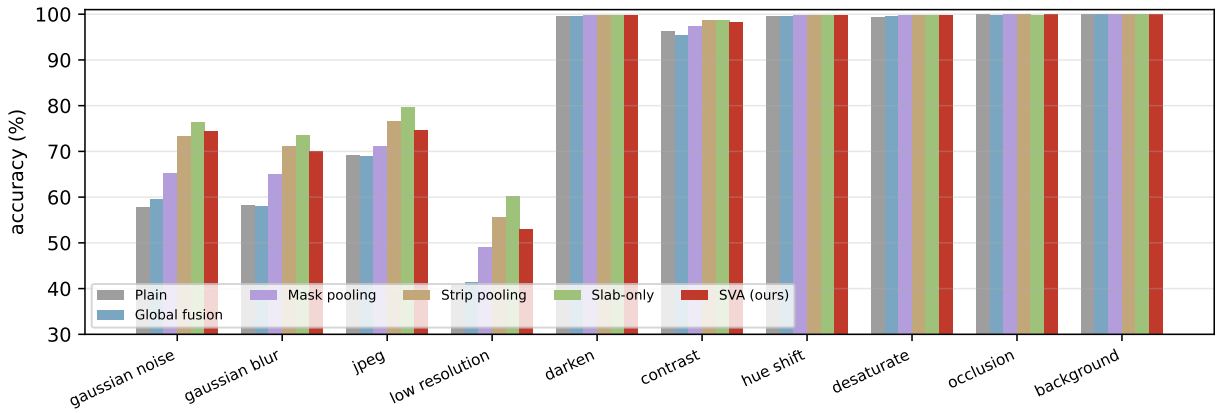


Figure 4: MedLeaf-C accuracy per corruption family (ConvNeXt-T, severity-averaged).

i.e. from degraded texture; colour, occlusion and background corruptions are near ceiling for all models. On the other backbones all variants are within about one point. Shape bias remains low everywhere (7–14%): when a silhouette is filled with another species' lamina, all models mostly follow the lamina. Localised pooling raises ConvNeXt-T shape bias only slightly (plain 6.5%, strip 9.1%, SVA 9.0%), and BioCLIP 2 is the most shape-biased backbone (14%). The localised models are therefore not shape-based; they are less brittle under texture degradation and domain shift.

6.6 Inside SVA: ablations

Table 4 varies one ingredient of SVA at a time (ConvNeXt-T). A single-seed screen on folds 0–2 (lower block) was followed by the four most informative ablations on all 5 folds × 3 seeds (upper block), compared with SVA on the same runs. The screen overstated every effect, because its SVA reference happened to be strong (53.1% raw vs. 49.3% over all seeds). With three seeds, a coarser sweep costs 5.1 [1.1, 9.1] raw points (15/15) and unit cell weights 4.5 [−0.5, 9.5] (13/15); Euler tokens transfer slightly worse than area and perimeter (−2.6, 12/15, n.s.); and restricting attention to the slab is not needed (global attention 49.3% vs. 49.3%). These effects concern the internal design of SVA; as shown above, the full module is not better than localised pooling without any measurement. The vein head reproduces its ridge-filter supervision closely on held-out leaves (Pearson $r = 0.96$, Dice 0.74–0.75 inside the blade), which measures agreement with the pseudo-label, not with traced veins.

6.7 Topology sensitivity

The topology-destruction probe compares edits that change the blade's Euler characteristic (three interior holes, $\Delta\chi = -3$) with edits that remove the same area from the margin ($\Delta\chi = 0$). The two edit types are not perceptually identical (median LPIPS 0.045 vs. 0.030), so we compare them only within LPIPS strata that contain both (10 quantile bins, at least five edits of each kind) and report the stratum-weighted log-ratio of the prediction change (KL divergence), $\log(\mathrm{KL}_{\mathrm{holes}}/\mathrm{KL}_{\mathrm{bites}})$, with a bootstrap 95% interval (table 5). Every ImageNet ConvNeXt-T and ViT-S model responds *less* to holes than to margin bites, even though the holes are the more visible edit. The only ConvNeXt-T model that responds more to holes is the Euler-token variant of SVA (+0.73 [+0.54, +0.96]), so measured topology does change what a model is sensitive to. BioCLIP 2 responds more to holes for every variant (+1.3 to +1.6), i.e. the foundation model is topology-sensitive by itself. Topological

Table 4: Ablations of SVA (ConvNeXt-T, %), each block sorted by raw field accuracy and compared with SVA on the same runs. Rotation is evaluated on seed 0 only.

Variant	Clean	Rot. min	Field (mask)	Field (raw)
5 folds × seeds 0–2				
SVA (full)	100.0 ± 0.1	99.9 ± 0.1	43.1 ± 2.8	49.3 ± 2.8
attention over all leaf patches	100.0 ± 0.0	100.0 ± 0.0	44.0 ± 2.6	49.3 ± 2.4
Euler token	100.0 ± 0.0	100.0 ± 0.0	42.6 ± 1.4	46.8 ± 2.1
unit cell weights	100.0 ± 0.1	99.8 ± 0.2	41.3 ± 1.4	44.8 ± 1.6
coarse sweep (K=16, T=8)	100.0 ± 0.1	99.9 ± 0.2	41.5 ± 2.1	44.2 ± 3.2
seed 0, folds 0–2				
SVA (full)	99.9 ± 0.2	99.9 ± 0.2	47.0 ± 0.7	53.1 ± 2.7
softer relaxation (κ=1)	100.0 ± 0.0	99.9 ± 0.2	46.8 ± 1.1	52.5 ± 2.1
attention over all leaf patches	100.0 ± 0.0	100.0 ± 0.0	47.1 ± 1.1	52.0 ± 2.5
weights permuted within the leaf	100.0 ± 0.0	100.0 ± 0.0	44.5 ± 0.8	50.8 ± 1.1
venation field only	99.9 ± 0.2	99.8 ± 0.2	45.3 ± 0.6	50.7 ± 2.5
sharper relaxation (κ=16)	100.0 ± 0.0	99.9 ± 0.2	46.5 ± 2.6	50.6 ± 3.8
blade only	100.0 ± 0.0	99.9 ± 0.2	43.8 ± 2.0	48.4 ± 4.1
coarse sweep (K=16, T=8)	100.0 ± 0.0	99.9 ± 0.2	43.2 ± 1.2	46.8 ± 1.0
Euler token	100.0 ± 0.0	100.0 ± 0.0	43.0 ± 0.7	45.7 ± 2.8
Hadwiger token (area, perimeter, Euler)	100.0 ± 0.0	100.0 ± 0.0	38.9 ± 1.1	44.3 ± 1.5
unit cell weights	99.9 ± 0.2	99.8 ± 0.2	41.5 ± 1.2	43.5 ± 0.7

sensitivity can thus be induced, but it did not help species transfer (section 6.6). Edits of the venation could not be matched perceptually (only 25% of vein edits fall inside the LPIPS range of their controls), so we draw no conclusion from them.

6.8 Leakage audit

Under random image-level splits that keep the duplicates, 5–11 test images per fold (1.4–3.0%) have an exact duplicate or a re-shot of the same leaf in the training part; under our grouped protocol there are none. On this dataset the leak does not measurably inflate accuracy: a DeepHerb-style reproduction (Xception features with an RBF-SVM [2]) reaches 98.5 ± 0.9% with random splits and 98.4 ± 0.7% with grouped splits, the dual-branch RGB + vein-skeleton baseline [8] reaches 99.95% under both, and our plain and SVA models reach 100% under random splits as under grouped ones. We therefore present the grouped protocol as hygiene that removes a known leak rather than as evidence that earlier results were inflated; the saturation itself is the more important observation.

7 Discussion

What closes the studio-to-field gap. For an ImageNet CNN trained on studio leaves, two simple properties of the pooling account for the whole architectural gain we observed: knowing where the leaf is, and pooling locally within it. Each is worth 5–8 points on raw field photographs, they add up to 12–14 points, and both survive without rotation augmentation and on top of strong background-compositing augmentation. Background compositing and mask pooling act similarly (each lifts plain ConvNeXt-T to about 41–42%), which supports the reading that the first part is about discounting the background. The second part, localisation, helps on raw photographs and

Table 5: LPIPS-matched response to topology-changing holes relative to area-matched margin bites, $\log(\mathrm{KL}_{\mathrm{holes}}/\mathrm{KL}_{\mathrm{bites}})$ with bootstrap 95% CI (seed 0, 5 folds). Negative: the model reacts more to margin geometry than to topology.

Backbone	Model	log-ratio [95% CI]
ConvNeXt-T	Plain	-1.22 [-1.47, -1.02]
	Global fusion	-0.95 [-1.15, -0.75]
	Strip pooling	-0.90 [-1.02, -0.77]
	Slab-only	-0.96 [-1.11, -0.81]
	SVA	-1.63 [-1.76, -1.48]
	Euler token (ablation)	+0.73 [+0.54, +0.96]
ViT-S/16	Plain	-0.35 [-0.51, -0.22]
	Global fusion	-0.62 [-0.76, -0.49]
	Strip pooling	-0.84 [-1.03, -0.65]
	Slab-only	-0.70 [-0.88, -0.52]
	SVA	-0.42 [-0.60, -0.25]
BioCLIP 2	Plain	+1.63 [+1.50, +1.76]
	Global fusion	+1.47 [+1.34, +1.60]
	Strip pooling	+1.34 [+1.23, +1.45]
	Slab-only	+1.36 [+1.23, +1.50]
	SVA	+1.60 [+1.46, +1.74]

on texture corruptions but hardly on whitened photographs, so it seems to make features robust to clutter and texture loss inside and around the leaf rather than to add shape information; shape bias barely changes.

What does not. We expected measured leaf geometry to matter: an object-centred sweep, per-slab area and boundary length of the blade and venation, and attention restricted to the slab being measured. None of these ingredients improves transfer beyond localised pooling. Object-centred slabs are no better than image-axis strips, the same measurements fused globally give nothing, and restricting attention to the measured slab can be replaced by global attention. Within SVA, a fine sweep and learned weights matter, but only relative to weaker versions of SVA itself. Euler tokens make a model sensitive to holes in the blade, yet transfer worse. Within the limits of our design, the effect we can detect is about *where* features are pooled, not about measuring the leaf. We report SVA because a well-controlled negative result for a natural and increasingly popular idea, fusing topological and geometric measurements with deep features, is useful, and because its localised variant is among the best models we trained.

Practical recommendations. (i) Evaluate leaf classifiers on field photographs; studio accuracy is saturated and uninformative (98–100% for every model). (ii) With an ImageNet CNN, pool over the leaf and locally within it, e.g. with strip pooling, which is simpler and cheaper than SVA (section D), and combine it with background-compositing augmentation; together they raised field accuracy from 34% to 53–55%. (iii) Where a biology foundation model such as BioCLIP 2 can be used, use it: it reaches 81% without any module, and none of the modules changes it.

Limitations. (i) One studio dataset and one field dataset with 20 shared species. Field labels come from vernacular folder names that we mapped to species and checked visually; residual label

noise would lower all models alike. The 20-way accuracy uses the knowledge of which species can occur. (ii) The masked field condition and all localised models depend on our masker, whose accuracy on field photographs we could not measure without ground-truth masks; the raw-photo results, which the plain model obtains without any mask, are the primary outcome. (iii) ConvNeXt-T results use five seeds, EfficientNet-B0 and the key ablations three, ViT-S and BioCLIP 2 one; differences of 2–3 points between localised models are not resolved by our design, and the ViT results are indicative only. (iv) One fixed training recipe was used for all models of a backbone; a recipe tuned per model could shift individual numbers. (v) The venation field is learned from ridge-filter pseudo-labels and is not validated against traced veins.

8 Conclusion

Medicinal-leaf classifiers that are perfect on studio photographs transfer poorly to field photographs. In a controlled study with matched models, five seeds and corrected tests, we find that for an ImageNet CNN the architectural part of the remedy is simple: pool over the leaf, and pool locally within it. Each step is worth 5–8 points on field photographs, and both add to background-compositing augmentation. Measured leaf geometry, introduced here as sweep-valuation attention with appearance-weighted area, boundary and Euler measurements along an object-centred sweep, does not improve transfer beyond localised pooling, and a biology foundation model makes all such modules unnecessary. We release the leakage-audited protocol, the field mapping, the corruption, cue-conflict and topology probes, and all code and run outputs.

Code and data availability

All code, configurations, unit tests, the outputs of every training run (metrics, probe results, per-image predictions and logs), the statistical analyses and the LaTeX sources of this paper are publicly available at `https://github.com/Devguru-codes/research_paper_leaf`. Every number reported here is regenerated from those run outputs by the scripts in the repository, and the properties stated in section 4 are verified by its unit tests (section A). The image datasets are not redistributed; they are available from their original sources (section 3) [32, 33].

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A Unit tests of the structural signals

All properties used in section 4 are checked by automated tests that ship with the code (35 tests, all passing; the list covers the properties used in this paper).

- **Exact Euler characteristic.** On toy shapes (disk, annulus, two disks, a lens-shaped leaf, and the lens with two holes; $\chi = 1, 0, 2, 1, -1$) the cubical $\chi = V - E + F$ of the binary set equals the value computed independently with scikit-image (4-connected foreground).
- **Telescoping.** For every direction, $\sum_k \Delta\chi(\theta, k)$, $\sum_k \Delta A(\theta, k)$ and $\sum_k \Delta P(\theta, k)$ equal the total valuation of the shape, to numerical precision.
- **Partition of unity.** Slab memberships (4) sum to one for every height.
- **Sharp limit.** With binary weights and large κ, the relaxed increments equal those of a hard sweep that assigns each cell to the slab containing its height.
- **Closed form.** For an $n \times n$ block of pixels, the totals are $(A, P, \chi) = ((n-1)^2, 4(n-1), 1)$.
- **Consistency.** The Euler column of the Hadwiger increments equals the separately implemented Euler increments.
- **Rotation.** For unit weights, rotating the input by 90° cyclically shifts $\Delta\chi$, ΔA and ΔP by $K/4$ directions. This follows from $\langle \mathcal{R}x, \theta \rangle = \langle x, \mathcal{R}^{-1}\theta \rangle$, the equivariance of the centroid/RMS frame and the fact that a 90° grid rotation maps cells of each dimension bijectively onto cells of the same dimension; for rotations by $2\pi/K$ it holds up to grid resampling.
- **Mask pooling.** The mask-pooling control is unchanged when patch tokens outside the leaf mask are altered.
- **Fuzzy subcomplex.** Learned cell weights satisfy $w(\text{coface}) \leq w(\text{face})$, and gradients reach the per-pixel weights.
- **Splits.** Under the grouped protocol no exact duplicate or re-shot group crosses a fold boundary and every image is tested exactly once; under the random protocol leaks occur.

B Species and taxonomy

C Field-photo test set

D Computational cost

Table 6: The 30 species (APG IV order and family) and the number of images after de-duplication.

Order / family	Species	Name	n
Brassicales / Brassicaceae	*Brassica juncea*	Indian Mustard	34
Brassicales / Moringaceae	*Moringa oleifera*	Drumstick	77
Caryophyllales / Amaranthaceae	*Amaranthus viridis*	Arive-Dantu	122
Caryophyllales / Basellaceae	*Basella alba*	Basale	103
Fabales / Fabaceae	*Pongamia pinnata*	Indian Beech	61
Fabales / Fabaceae	*Trigonella foenum-graecum*	Fenugreek	36
Gentianales / Apocynaceae	*Carissa carandas*	Karanda	64
Gentianales / Apocynaceae	*Nerium oleander*	Oleander	62
Gentianales / Apocynaceae	*Tabernaemontana divaricata*	Crape Jasmine	56
Lamiales / Lamiaceae	*Mentha sp.*	Mint	97
Lamiales / Lamiaceae	*Ocimum tenuiflorum*	Tulsi	52
Lamiales / Lamiaceae	*Plectranthus amboinicus*	Mexican Mint	48
Lamiales / Oleaceae	*Jasminum sp.*	Jasmine	71
Lamiales / Oleaceae	*Nyctanthes arbor-tristis*	Parijata	40
Malvales / Malvaceae	*Hibiscus rosa-sinensis*		43
Malvales / Muntingiaceae	*Muntingia calabura*	Jamaica Cherry-Gasagase	56
Myrtales / Lythraceae	*Punica granatum*	Pomegranate	79
Myrtales / Myrtaceae	*Psidium guajava*	Guava	65
Myrtales / Myrtaceae	*Syzygium cumini*	Jamun	39
Myrtales / Myrtaceae	*Syzygium jambos*	Rose Apple	56
Piperales / Piperaceae	*Piper betle*	Betel	48
Rosales / Moraceae	*Artocarpus heterophyllus*	Jackfruit	56
Rosales / Moraceae	*Ficus auriculata*	Roxburgh fig	50
Rosales / Moraceae	*Ficus religiosa*	Peepal Tree	63
Santalales / Santalaceae	*Santalum album*	Sandalwood	58
Sapindales / Anacardiaceae	*Mangifera indica*	Mango	62
Sapindales / Meliaceae	*Azadirachta indica*	Neem	60
Sapindales / Rutaceae	*Citrus limon*	Lemon	57
Sapindales / Rutaceae	*Murraya koenigii*	Curry	60
Zingiberales / Zingiberaceae	*Alpinia galanga*	Rasna	50

Table 7: DIMPSAR folders used for the field test, the species they were mapped to (checked visually), and the number of images in which our masker found a leaf. Kannada *Arali* denotes *Ficus religiosa*, not *Nerium oleander*.

DIMPSAR folder	Species	Family	Images used
Jackfruit	*Artocarpus heterophyllus*	Moraceae	110
Neem	*Azadirachta indica*	Meliaceae	113
Malabar Spinach	*Basella alba*	Basellaceae	79
Lemon	*Citrus limon*	Rutaceae	120
Arali	*Ficus religiosa*	Moraceae	89
Hibiscus	*Hibiscus rosa-sinensis*	Malvaceae	113
Jasmine	*Jasminum sp.*	Oleaceae	48
Mango	*Mangifera indica*	Anacardiaceae	103
Mint	*Mentha sp.*	Lamiaceae	122
Drumstick	*Moringa oleifera*	Moringaceae	26
Gasagase	*Muntingia calabura*	Muntingiaceae	77
Curry	*Murraya koenigii*	Rutaceae	159
Parijatha	*Nyctanthes arbor-tristis*	Oleaceae	61
Tulsi	*Ocimum tenuiflorum*	Lamiaceae	171
Betel	*Piper betle*	Piperaceae	112
Doddpathre	*Plectranthus amboinicus*	Lamiaceae	131
Honge	*Pongamia pinnata*	Fabaceae	110
Guava	*Psidium guajava*	Myrtaceae	121
Pomoegranate	*Punica granatum*	Lythraceae	73
Nerale	*Syzygium cumini*	Myrtaceae	62

Table 8: Per-species 20-way accuracy (%) on raw field photographs, ConvNeXt-T, mean over all runs.

Species	n	Plain	Global	Strip	Slab-only	SVA
Syzygium cumini	62	15	12	11	4	2
Nyctanthes arbor-tristis	61	6	5	3	10	5
Jasminum sp.	48	2	0	8	7	7
Pongamia pinnata	110	13	5	24	14	18
Mangifera indica	103	12	16	22	18	21
Moringa oleifera	26	33	35	33	28	33
Artocarpus heterophyllus	110	24	21	34	33	35
Mentha sp.	122	20	24	53	56	47
Murraya koenigii	159	59	52	56	44	47
Citrus limon	120	10	16	23	29	50
Hibiscus rosa-sinensis	113	30	33	42	43	53
Ocimum tenuiflorum	171	41	35	54	51	53
Psidium guajava	121	45	38	51	54	55
Azadirachta indica	113	16	23	44	34	57
Punica granatum	73	16	39	73	73	62
Plectranthus amboinicus	131	38	36	71	70	63
Muntingia calabura	77	31	29	57	58	67
Basella alba	79	56	54	63	63	67
Piper betle	112	78	82	85	81	87
Ficus religiosa	89	80	84	93	90	95

Table 9: Parameters, GFLOPs (one image at training resolution) and median batch-1 latency on a laptop RTX 3050 GPU and on CPU. Mask pooling and strip pooling add almost no cost; the sweep-based models roughly double CPU and GPU latency, mostly because of the histogram operations.

Backbone	Model	Params (M)	Added (M)	GFLOPs	GPU ms	CPU ms
ConvNeXt-T	Plain	27.9	0.12	27.9	16	341
	Mask pooling	28.1	0.23	28.0	21	347
	Strip pooling	28.9	1.11	28.0	18	318
	Slab-only	29.1	1.26	29.3	26	661
	SVA	29.1	1.30	29.3	30	621
	Global fusion	29.6	1.79	28.1	28	411
EfficientNet-B0	Plain	4.1	0.09	3.2	23	270
	Strip pooling	5.0	1.03	3.2	25	294
	Slab-only	5.1	1.12	4.3	31	386
	SVA	5.2	1.16	4.3	49	431
	Global fusion	5.8	1.75	3.3	39	348
ViT-S/16	Plain	22.1	0.27	29.6	16	462
	Strip pooling	23.1	1.27	29.7	17	510
	Slab-only	23.2	1.42	31.2	22	479
	SVA	23.3	1.46	31.3	30	635
	Global fusion	23.8	1.96	30.1	32	569